



HORMONES, SEXUAL HEALTH & MISC

Oxytocin Acetate

Oxytocin Acetate – 2 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Oxytocin (acetate salt)
SYNONYMS / ALIASES	Oxytocin; OXT; alpha-hypophamine
CAS NUMBER	50-56-6 (free peptide); 6233-83-6 (acetate salt)
MOLECULAR FORMULA	C43H66N12O12S2 (free peptide)
MOLECULAR WEIGHT	1007.19 g/mol (free peptide; acetate salt mass varies with counterion ratio)
AMINO ACID SEQUENCE	H-Cys-Tyr-Ile-Gln-Asn-Cys-Pro-Leu-Gly-NH2 (9 residues; disulfide bond between Cys1 and Cys6 forming a 6-residue intramolecular ring; C-terminal glycynamide)
SOURCE	Synthetic (solid-phase peptide synthesis with oxidative disulfide formation); sequence corresponds to native mammalian oxytocin

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	2 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Disulfide bond integrity is critical; avoid reducing agents and prolonged exposure to elevated pH or temperature. The C-terminal glycynamide is generally stable.

STRUCTURAL NOTES

Oxytocin is a 9-amino-acid cyclic neuropeptide whose sequence is conserved across mammals. The molecule contains an intramolecular disulfide bridge between cysteines at positions 1 and 6, creating a 6-residue ring with a 3-residue C-terminal tail terminating in a glycnamide. Oxytocin differs from the structurally related neuropeptide arginine vasopressin (AVP) at only two positions (3 and 8): oxytocin contains isoleucine and leucine where AVP contains phenylalanine and arginine. The acetate salt form is the most common counterion presentation.

SELECTED PEER-REVIEWED REFERENCES

1. du Vigneaud V. et al. - *The synthesis of an octapeptide amide with the hormonal activity of oxytocin* - Journal of the American Chemical Society, 1953 - DOI: 10.1021/ja01115a503
2. Gimpl G., Fahrenholz F. - *The oxytocin receptor system: structure, function, and regulation* - Physiological Reviews, 2001 - DOI: 10.1152/physrev.2001.81.2.629
3. Lee H.J. et al. - *Oxytocin: the great facilitator of life* - Progress in Neurobiology, 2009 - DOI: 10.1016/j.pneurobio.2009.04.001

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.